Economia das Organizações e dos Incentivos — MPE Insper

Exercise List: Real Book Exercises, Lectures 1–5

How to use this list. Every exercise below is a real exercise from one of the course books, reproduced verbatim: the source line gives author, book, chapter, exercise, the printed page and, in brackets, the page your PDF viewer shows for the file in `Livros/`, and whether the source gives the answer. Muñoz-Garcia & Toro-Gonzalez print a worked answer right after each exercise; MWG answers are in the MWG solutions manual in `Livros/`; Tadelis, Gibbons and Cabral give no answer in the book. When an exercise refers to a model, game or figure stated elsewhere in its book, that material is copied into the same block under "Referenced material", so every exercise can be solved where it is printed. There is one block per lecture of the real schedule. Each block has ten exercises in strictly rising difficulty, rated from 1 (apply a definition) to 5 (graduate level) next to each heading: three foundations, five exam level, two advanced. Try each exercise in the space provided before opening the book's answer.

Difficulty path (what makes each block's last items hard):

- **Lecture 1**: from one 3×3 elimination (1) through two short proofs about dominance (Tadelis 4.1, MWG 8.B.4) to MWG 8.B.5 (5), where elimination over a continuum of quantities must be shown to converge with two firms and to stall with three.
- **Lecture 2**: from writing a matrix (1) through the Bertrand uniqueness proof and MWG 8.D.2 to MWG 8.D.6 (5), a proof that every 2×2 game, with arbitrary payoffs, has a mixed-strategy equilibrium.
- **Lecture 3**: from counting strategies (1) through Nature, imperfect recall and the zero-sum proof (MWG 9.B.4) to Gibbons 2.1 (5), a proof with general concave income and utility functions and continuous actions.
- **Lecture 4**: from definitions (1) through entry, Stackelberg, hold-up and the legal-system game (Tadelis 8.7) to MWG 9.B.5 (5), a five-part general-payoff argument on strategy counts, multiple SPNE and when moving first guarantees the simultaneous-move payoff, with counterexamples.
- **Lecture 5**: from unraveling (1) through punishments of finite length, carrot and stick and the both-horizons bar game (Tadelis 10.7) to Gibbons 2.20 (5), a proof that Rubinstein's stationary strategies are a Nash equilibrium and are subgame-perfect, checking every history of an infinite game.

Selection tables (one per lecture, next pages).

Items not in the list for Lectures 1–4 (`lista-reais-lectures-01-04.tex`) are marked: † added in the 30-item version, ‡ added in the 50-item version. Every other item is reused from that list with its statement unchanged, some in a different block or position. "Diff.": difficulty from 1 (apply a definition) to 5 (graduate level). Pages: printed page, then the PDF viewer page in brackets (offsets in `Livros/book-registry.json`; MWG is a two-pages-per-sheet scan). "Answer": *book* = worked answer printed after the exercise; *manual* = MWG solutions manual in `Livros/`; *none* = no answer in the book. Books: Tadelis (2013), *Game Theory: An Introduction*; Gibbons (1992), *Game Theory for Applied Economists*; Muñoz-Garcia & Toro-Gonzalez (2019), *Strategy and Game Theory: Practice Exercises with Answers*, 2nd ed.; Mas-Colell, Whinston & Green (1995), *Microeconomic Theory* (MWG); Cabral (2017), *Introduction to Industrial Organization*, 2nd ed.

Block	#	Source	p. [PDF]	Level	What it trains	Diff.	Answer
L1	1	Tadelis 4.5	77 [96]	Found.	IESDS round by round, 3×3 matrix	1	none
	2	Muñoz-Garcia Ch. 1 Ex. 13‡	26 [39]	Found.	Dominant and dominated strategies, then IDSDS	1	book
	3	Gibbons 1.2	48 [56]	Found.	IESDS, then pure Nash equilibria	2	none
	4	Tadelis 4.1‡	76 [95]	Exam	Proof: a strictly dominant-strategy equilibrium is unique	2	none
	5	Muñoz-Garcia Ch. 1 Ex. 12	23 [36]	Exam	Rationality vs. common knowledge of it	2	book
	6	Tadelis 4.7	77–78 [96–97]	Exam	Story to matrix, then elimination	3	none
	7	Tadelis 4.3‡	76 [95]	Exam	Discrete first-price auction: matrix, IESDS	3	none
	8	Tadelis 4.6	77 [96]	Exam	IESDS with continuous strategies	4	none
	9	MWG 8.B.4‡	262 [142]	Adv.	Proof: the order of strict deletion does not matter	4	manual
	10	MWG 8.B.5	262 [142]	Adv.	Cournot by IESDS: two firms vs. three	5	manual

Block	#	Source	p. [PDF]	Level	What it trains	Diff.	Answer
L2	11	Tadelis 5.6	96 [115]	Found.	Hawk-dove: matrix, pure equilibria	1	none
	12	Muñoz-Garcia Ch. 2 Ex. 2‡	41 [54]	Found.	Battle of the sexes: pure equilibria	1	book
	13	Muñoz-Garcia Ch. 3 Ex. 1	86–87 [99–100]	Found.	Chicken: pure and mixed equilibria	2	book
	14	Gibbons 1.12‡	51 [59]	Exam	A mixed equilibrium by indifference	2	none
	15	Tadelis 5.8	96–97 [115–116]	Exam	Cournot with n firms; competitive limit	3	none
	16	Gibbons 1.6	49 [57]	Exam	Cournot, asymmetric costs; corner case	3	none
	17	Gibbons 1.7‡	49–50 [57–58]	Exam	Homogeneous Bertrand: $p_1 = p_2 = c$ is the unique equilibrium	3	none
	18	Tadelis 6.8	125 [144]	Exam	Three-player entry; symmetric mixing	4	none
	19	MWG 8.D.2‡	263 [142]	Adv.	Proof: a unique IESDS survivor is a Nash equilibrium	4	manual
	20	MWG 8.D.6	264 [143]	Adv.	A mixed equilibrium exists in every 2×2 game	5	manual

Block	#	Source	p. [PDF]	Level	What it trains	Diff.	Answer
L3	21	Tadelis 7.1‡	147 [166]	Found.	Counting pure strategies over information sets	1	none
	22	Muñoz-Garcia Ch. 4 Ex. 3	160 [172]	Found.	Sequential prisoner's dilemma	1	book
	23	Tadelis 7.2†	147 [166]	Found.	Story to tree; strategies; an "unappealing" Nash equilibrium	2	none
	24	Tadelis 8.5†	170 [189]	Exam	A simultaneous game behind an adoption choice: SPE	3	none
	25	Tadelis 8.10	172 [191]	Exam	Subgames with a simultaneous stage	3	none
	26	Tadelis 7.4‡	148 [167]	Exam	Four-move centipede: strategies, Nash equilibria, outcomes	3	none
	27	Tadelis 7.9‡	149–150 [168–169]	Exam	A move of Nature and an information set	4	none
	28	Tadelis 7.10†	150 [169]	Exam	Mixed vs. behavioral strategies; perfect recall	4	none
	29	MWG 9.B.4‡	302 [162]	Adv.	Proof: unique SPNE payoffs in zero-sum games of perfect information	4	manual
	30	Gibbons 2.1	130 [137]	Adv.	Rotten Kid theorem (continuous actions)	5	none

Block	#	Source	p. [PDF]	Level	What it trains	Diff.	Answer
L4	31	Gibbons 2.12†	135 [142]	Found.	Strategy, subgame and SPE in a repeated game	1	none
	32	Tadelis 7.6‡	148–149 [167–168]	Found.	Entry game: payoffs from preferences, all Nash equilibria	1	none
	33	Tadelis 8.8	171–172 [190–191]	Found.	Entry deterrence: a non-credible threat	2	none
	34	Tadelis 8.6†	170–171 [189–190]	Exam	Sunk investment before Cournot; threshold F^*	3	none
	35	Muñoz-Garcia Ch. 4 Ex. 11‡	180 [192]	Exam	Stackelberg, asymmetric costs; vs. monopoly, Cournot, Bertrand	3	book
	36	Tadelis 8.11‡	172–173 [191–192]	Exam	Sequential vetoes; Nash equilibria that are not SPE; removing an option	3	none
	37	Tadelis 8.4†	170 [189]	Exam	Leader, two followers: subgames; a Nash equilibrium that is not SPE	4	none
	38	Tadelis 8.13	173–174 [192–193]	Exam	Hold-up; a posted price as commitment	4	none
	39	Tadelis 8.7‡	171 [190]	Adv.	Credibility of suing; design of the legal system	4	none
	40	MWG 9.B.5†	302 [162]	Adv.	Observation, multiple SPNE, when moving first pays	5	manual

Block	#	Source	p. [PDF]	Level	What it trains	Diff.	Answer
L5	41	Cabral 7.13†	182 [194]	Found.	Finitely repeated PD unravels	1	none
	42	Tadelis 10.2†	215 [234]	Found.	Grim trigger; threshold δ for two paths	2	none
	43	Muñoz-Garcia Ch. 6 Ex. 1‡	274 [285]	Found.	One-, two- and ten-period punishments	2	book
	44	Muñoz-Garcia Ch. 6 Ex. 9†	305 [316]	Exam	Feasible, individually rational payoffs; folk theorem	3	book
	45	Tadelis 10.3‡	215–216 [234–235]	Exam	Length-T punishments: δ_1 vs. δ_2	3	none
	46	Tadelis 9.1	186–187 [205–206]	Exam	Stage game played twice: all SPE, carrot and stick	4	none
	47	Gibbons 2.11‡	135 [142]	Exam	A non-Nash first-stage payoff in a twice-played game	4	none
	48	Tadelis 11.5	236–237 [255–256]	Exam	Three periods with $\delta_1 \neq \delta_2$	4	none
	49	Tadelis 10.7‡	217–218 [236–237]	Adv.	Quality: ten days vs. forever, in parameters	4	none
	50	Gibbons 2.20†	137–138 [144–145]	Adv.	Rubinstein strategies are an SPE (one-stage deviations)	5	none

Block 1 — Static games I: dominance and iterated elimination *(Lecture 1)*

Exercise 1. *Iterated Elimination* [Foundations; difficulty 1/5]

Source. Tadelis (2013), *Game Theory: An Introduction*, Ch. 4 "Rationality and Common Knowledge", Exercise 4.5 "Iterated Elimination", p. 77 [PDF 96]. No answer in the book. Reproduced verbatim.

4.5 Iterated Elimination: In the following normal-form game, which strategy profiles survive iterated elimination of strictly dominated strategies?

		Player 2		
		L	*C*	*R*
Player 1	*U*	6, 8	2, 6	8, 2
	M	8, 2	4, 4	9, 5
	D	8, 10	4, 6	6, 7

Resolution:

Exercise 2. *Iterated Deletion of Strictly Dominated Strategies (IDSDS)* [Foundations; difficulty 1/5]

Source. Muñoz-Garcia & Toro-Gonzalez (2019), *Strategy and Game Theory: Practice Exercises with Answers*, 2nd ed., Ch. 1 "Dominance Solvable Games", Exercise 13 "Iterated Deletion of Strictly Dominated Strategies (IDSDS)"[A], p. 26 [PDF 39]. Worked answer in the book, right after the statement. Reproduced verbatim (statement only); Fig. 1.33, printed inside the answer on the same page, rebuilt.

Consider the following normal form game (Fig. 1.33)

(a) Find strictly dominant strategies (if any) for player 1 and for player 2.

(b) Find strictly dominated strategies (if any) for player 1 and for player 2.

(c) If you apply iterative deletion of strictly dominated strategies (IDSDS), what is the surviving strategy pair (or pairs)? Explain the steps you use in IDSDS, and why you use them.

		Player 2	
		Left	*Right*
Player 1	*Up*	-10, 10	0, 12
	Middle	-12, 0	2, 2
	Down	-9, 0	1, 0

Fig. 1.33 Normal-form game

Resolution:

Exercise 3. *IESDS and pure-strategy Nash equilibria* [Foundations; difficulty 2/5]

Source. Gibbons (1992), *Game Theory for Applied Economists*, Ch. 1 "Static Games of Complete Information", §1.5 Problems (Section 1.1), Problem 1.2, p. 48 [PDF 56]. No answer in the book. Reproduced verbatim; the payoff pair in cell (T, L) is printed "2.0" (comma typo) and is typeset "2, 0".

1.2. In the following normal-form game, what strategies survive iterated elimination of strictly dominated strategies? What are the pure-strategy Nash equilibria?

	L	C	R
T	2, 0	1, 1	4, 2
M	3, 4	1, 2	2, 3
B	1, 3	0, 2	3, 0

Resolution:

Exercise 4. *Prove Proposition 4.1* [Exam level; difficulty 2/5]

Source. Tadelis (2013), *Game Theory: An Introduction*, Ch. 4 "Rationality and Common Knowledge", Exercise 4.1, p. 76 [PDF 95]. No answer in the book. Reproduced verbatim. The exercise states the proposition it asks to prove.

4.1 Prove Proposition 4.1: If the game $\Gamma = \langle N, \{S_i\}_{i=1}^n, \{v_i\}_{i=1}^n \rangle$ has a strictly dominant strategy equilibrium s^D, then s^D is the unique dominant strategy equilibrium.

Resolution:

Exercise 5. *IDSDS and Rationality* [Exam level; difficulty 2/5]

Source. Muñoz-Garcia & Toro-Gonzalez (2019), *Strategy and Game Theory: Practice Exercises with Answers*, 2nd ed., Ch. 1 "Dominance Solvable Games", Exercise 12 "IDSDS and Rationality"[A], p. 23 [PDF 36]. Worked answer in the book, right after the statement. Reproduced verbatim (statement only).

Consider the following 3×3 matrix (Fig. 1.30):

		Player 2		
		F	C	B
Player 1	F	0, 5	2, 4	2, 3
	C	2, 3	0, 5	3, 2
	B	5, 0	2, 4	2, 3

Fig. 1.30 Normal-form representation of a three-players game

a. Does any player have a strictly dominant strategy?

b. Does any player have a strictly dominated strategy?

c. Use your results from part (b) to delete players' strictly dominated strategies assuming rationality, but not assuming common knowledge of rationality. [Hint: This means that every player can delete the strategies he finds to be strictly dominated but cannot put himself in the shoes of his opponent to delete the strategies his opponent finds to be strictly dominated.]

d. Use your results from part (b) to delete players' strictly dominated strategies assuming common knowledge of rationality. [Hint: This means that players can put themselves in the shoes of their opponents to delete strategies each other finds to be strictly dominated, as in the standard approach of IDSDS.]

Resolution:

Exercise 6. *Campaigning* [Exam level; difficulty 3/5]

Source. Tadelis (2013), *Game Theory: An Introduction*, Ch. 4 "Rationality and Common Knowledge", Exercise 4.7 "Campaigning", pp. 77–78 [PDF 96–97]. No answer in the book. Reproduced verbatim.

4.7 Campaigning: Two candidates, 1 and 2, are running for office. Each has one of three choices in running his campaign: focus on the positive aspects of one's own platform (call this a positive campaign $[P]$), focus on the positive aspects of one's own platform while attacking one's opponent's campaign (call this a balanced campaign $[B]$), or finally focus only on attacking one's opponent (call this a negative campaign $[N]$). All a candidate cares about is the probability of winning, so assume that if a candidate expects to win with probability $\pi \in [0, 1]$, then his payoff is π. The probability that a candidate wins depends on his choice of campaign and his opponent's choice. The probabilities of winning are given as follows:

If both choose the same campaign then each wins with probability 0.5.

If candidate i uses a positive campaign while $j \neq i$ uses a balanced one then i loses for sure.

If candidate i uses a positive campaign while $j \neq i$ uses a negative one then i wins with probability 0.3.

If candidate i uses a negative campaign while $j \neq i$ uses a balanced one then i wins with probability 0.6.

a. Model this story as a normal-form game. (It suffices to be specific about the payoff function of one player and to explain how the other player's payoff function is different and why.)

b. Write down the game in matrix form.

c. What happens at each stage of elimination of strictly dominated strategies? Will this procedure lead to a clear prediction?

Resolution:

Exercise 7. *Discrete First-Price Auction* [Exam level; difficulty 3/5]

Source. Tadelis (2013), *Game Theory: An Introduction*, Ch. 4 "Rationality and Common Knowledge", Exercise 4.3 "Discrete First-Price Auction", p. 76 [PDF 95]. No answer in the book. Reproduced verbatim.

4.3 Discrete First-Price Auction: An item is up for auction. Player 1 values the item at 3 while player 2 values the item at 5. Each player can bid either 0, 1, or 2. If player i bids more than player j then i wins the good and pays his bid, while the loser does not pay. If both players bid the same amount then a coin is tossed to determine who the winner is, and the winner gets the good and pays his bid while the loser pays nothing.

a. Write down the game in matrix form.

b. Does any player have a strictly dominated strategy?

c. Which strategies survive IESDS?

Resolution:

Exercise 8. *Roommates* [Exam level; difficulty 4/5]

Source. Tadelis (2013), *Game Theory: An Introduction*, Ch. 4 "Rationality and Common Knowledge", Exercise 4.6 "Roommates", p. 77 [PDF 96]. No answer in the book. Reproduced verbatim.

4.6 Roommates: Two roommates each need to choose to clean their apartment, and each can choose an amount of time $t_i \geq 0$ to clean. If their choices are t_i and t_j, then player i's payoff is given by $(10 - t_j)t_i - t_i^2$. (This payoff function implies that the more one roommate cleans, the less valuable is cleaning for the other roommate.)

a. What is the best response correspondence of each player i?

b. Which choices survive one round of IESDS?

c. Which choices survive IESDS?

Resolution:

Exercise 9. *Order of deletion* [Advanced; difficulty 4/5]

Source. Mas-Colell, Whinston & Green (1995), *Microeconomic Theory*, Ch. 8 "Simultaneous-Move Games", Exercises, Exercise 8.B.4^{C}, p. 262 [PDF 142, left half]. Answer in the MWG solutions manual (`Livros/`). Reproduced verbatim.

8.B.4^{C} Show that the order of deletion does not matter for the set of strategies surviving a process of iterated deletion of strictly dominated strategies.

Resolution:

Exercise 10. *Cournot by successive elimination* [Advanced; difficulty 5/5]

Source. Mas-Colell, Whinston & Green (1995), *Microeconomic Theory*, Ch. 8 "Simultaneous-Move Games", Exercises, Exercise 8.B.5^{C}, p. 262 [PDF 142, left half]. Answer in the MWG solutions manual (`Livros/`; 8.B.5 is there). Reproduced verbatim. The pointer to Chapter 12 is context; the statement defines the whole model.

8.B.5^{C} Consider the Cournot duopoly model (discussed extensively in Chapter 12) in which two firms, 1 and 2, simultaneously choose the quantities they will sell on the market, q_1 and q_2. The price each receives for each unit given these quantities is $P(q_1, q_2) = a - b(q_1 + q_2)$. Their costs are c per unit sold.

(a) Argue that successive elimination of strictly dominated strategies yields a unique prediction in this game.

(b) Would this be true if there were three firms instead of two?

Resolution:

Block 2 — Static games II: Nash equilibrium and mixed strategies *(Lecture 2)*

Exercise 11. *Hawk-Dove* [Foundations; difficulty 1/5]

Source. Tadelis (2013), *Game Theory: An Introduction*, Ch. 5 "Pinning Down Beliefs: Nash Equilibrium", Exercise 5.6 "Hawk-Dove", p. 96 [PDF 115]. No answer in the book. Reproduced verbatim, with the book's footnote 12.

5.6 Hawk-Dove: The following game has been widely used in evolutionary biology to understand how fighting and display strategies by animals could coexist in a population. For a typical Hawk-Dove game there are resources to be gained (e.g., food, mates, territories), denoted as v. Each of two players can choose to be aggressive, as Hawk (H), or compromising, as Dove (D). If both players choose H then they split the resources but lose some payoff from injuries, denoted as k. Assume that $k > \frac{v}{2}$. If both choose D then they split the resources but engage in some display of power that carries a display cost d, with $d < \frac{v}{2}$. Finally, if player i chooses H while j chooses D then i gets all the resources while j leaves with no benefits and no costs.

a. Describe this game in a matrix.

b. Assume that $v = 10$, $k = 6$, and $d = 4$. What outcomes can be supported as pure-strategy Nash equilibria?[12]

[12] In the evolutionary biology literature, the analysis performed is of a very different nature. Instead of considering the Nash equilibrium analysis of a static game, the analysis is a dynamic one in which successful strategies "replicate" in a large population. This analysis is part of a methodology called evolutionary game theory. For more on the subject see Gintis (2000).

Resolution:

Exercise 12. *Battle of the Sexes* [Foundations; difficulty 1/5]

Source. Muñoz-Garcia & Toro-Gonzalez (2019), *Strategy and Game Theory: Practice Exercises with Answers*, 2nd ed., Ch. 2 "Pure Strategy Nash Equilibrium and Simultaneous-Move Games with Complete Information", Exercise 2 "Battle of the Sexes"[A], p. 41 [PDF 54]. Worked answer in the book, right after the statement. Reproduced verbatim (statement only); Fig. 2.4, printed inside the answer on the same page, rebuilt, with the book's row label "Housband" kept as printed.

A husband and a wife are leaving work, and do not remember which event they are attending to tonight. Both of them, however, remember that last night's argument was about either attending to the football game (the most preferred event for the husband) or the opera (the most preferred event for the wife). To make matters worse, their cell phones broke, so they cannot call each other to confirm which event they are attending to. As a consequence, they must simultaneously and independently decide whether to attend to the football game or the opera.

The payoff matrix in Fig. 2.4 describes the preference of the husband (wife) for the football game (opera, respectively), but also indicates that both players prefer to be together rather than being alone (even if they are alone at their most preferred event). Find the set of Nash Equilibria of this game.

		Wife	
		Football	*Opera*
Housband	*Football*	3, 1	0, 0
	Opera	0, 0	1, 3

Fig. 2.4 Battle of the sexes game (Normal-form representation)

Resolution:

Exercise 13. *Game of Chicken* [Foundations; difficulty 2/5]

Source. Muñoz-Garcia & Toro-Gonzalez (2019), *Strategy and Game Theory: Practice Exercises with Answers*, 2nd ed., Ch. 3 "Mixed Strategies, Strictly Competitive Games, and Correlated Equilibria", Exercise 1 "Game of Chicken"[A], pp. 86–87 [PDF 99–100]. Worked answer in the book, right after the statement. Reproduced verbatim (statement only).

Consider the game of Chicken (depicted in Fig. 3.1), in which two players driving their cars against each other must decide whether or not to swerve.

Part (a) Is there any strictly dominated strategy for Player 1? And for Player 2?

Part (b) What are the best responses for Player 1? And for Player 2?

Part (c) Can you find any pure strategy Nash Equilibrium (psNE) in this game?

Part (d) Find the mixed strategy Nash Equilibrium (msNE) of the game. *Hint*: denote by p the probability that Player 1 chooses Straight and by $(1-p)$ the probability that he chooses to Swerve. Similarly, let q denote the probability that Player 2 chooses Straight and $(1-q)$ the probability that she chooses to Swerve.

Part (e) Show your result of part (d) by graphically representing every player i's best response function $BRF_i(s_j)$, where $s_j = \{Swerve, Straight\}$ is the strategy selected by player $j \neq i$.

		Player 2	
		Straight	Swerve
Player 1	Straight	0,0	3,1
	Swerve	1,3	2,2

Fig. 3.1 Game of Chicken

Resolution:

Exercise 14. *A mixed equilibrium by indifference* [Exam level; difficulty 2/5]

Source. Gibbons (1992), *Game Theory for Applied Economists*, Ch. 1 "Static Games of Complete Information", §1.5 Problems (Section 1.3), Problem 1.12, p. 51 [PDF 59]. No answer in the book. Reproduced verbatim.

1.12. Find the mixed-strategy Nash equilibrium of the following normal-form game.

	L	R
T	2, 1	0, 2
B	1, 2	3, 0

Resolution:

Exercise 15. *The n-Firm Cournot Model* [Exam level; difficulty 3/5]

Source. Tadelis (2013), *Game Theory: An Introduction*, Ch. 5 "Pinning Down Beliefs: Nash Equilibrium", Exercise 5.8 "The n-Firm Cournot Model", pp. 96–97 [PDF 115–116]. No answer in the book. Reproduced verbatim (the printed "$Q = q_i + \cdots + q_n$" is kept as printed).

5.8 The n-Firm Cournot Model: Suppose there are n firms in the Cournot oligopoly model. Let q_i denote the quantity produced by firm i, and let $Q = q_i + \cdots + q_n$ denote the aggregate production. Let $P(Q)$ denote the market clearing price (when demand equals Q) and assume that the inverse demand function is given by $P(Q) = a - Q$, where $Q \leq a$. Assume that firms have no fixed cost and that the cost of producing quantity q_i is cq_i (all firms have the same marginal cost, and assume that $c < a$).

a. Model this as a normal-form game.

b. What is the Nash (Cournot) equilibrium of the game in which firms choose their quantities simultaneously?

c. What happens to the equilibrium price as n approaches infinity? Is this familiar?

Resolution:

Exercise 16. *Cournot with asymmetric costs* [Exam level; difficulty 3/5]

Source. Gibbons (1992), *Game Theory for Applied Economists*, Ch. 1 "Static Games of Complete Information", §1.5 Problems (Section 1.2), Problem 1.6, p. 49 [PDF 57]. No answer in the book. Reproduced verbatim.

1.6. Consider the Cournot duopoly model where inverse demand is $P(Q) = a - Q$ but firms have asymmetric marginal costs: c_1 for firm 1 and c_2 for firm 2. What is the Nash equilibrium if $0 < c_i < a/2$ for each firm? What if $c_1 < c_2 < a$ but $2c_2 > a + c_1$?

Referenced material. Source. "the Cournot duopoly model": Gibbons (1992), §1.2.A "Cournot Model of Duopoly", p. 15 [PDF 23], the paragraph that sets up the model. Reproduced verbatim; footnote 6 (a pointer to later sections) omitted.

Let q_1 and q_2 denote the quantities (of a homogeneous product) produced by firms 1 and 2, respectively. Let $P(Q) = a - Q$ be the market-clearing price when the aggregate quantity on the market is $Q = q_1 + q_2$. (More precisely, $P(Q) = a - Q$ for $Q < a$, and $P(Q) = 0$ for $Q \geq a$.) Assume that the total cost to firm i of producing quantity q_i is $C_i(q_i) = cq_i$. That is, there are no fixed costs and the marginal cost is constant at c, where we assume $c < a$. Following Cournot, suppose that the firms choose their quantities simultaneously.

Resolution:

Exercise 17. *Homogeneous Bertrand* [Exam level; difficulty 3/5]

Source. Gibbons (1992), *Game Theory for Applied Economists*, Ch. 1 "Static Games of Complete Information", §1.5 Problems (Section 1.2), Problem 1.7, pp. 49–50 [PDF 57–58]. No answer in the book. Reproduced verbatim. The pointer to Section 1.2.B (Bertrand with differentiated products) is context; the statement defines the whole homogeneous-products model.

1.7. In Section 1.2.B, we analyzed the Bertrand duopoly model with differentiated products. The case of homogeneous products yields a stark conclusion. Suppose that the quantity that consumers demand from firm i is $a - p_i$ when $p_i < p_j$, 0 when $p_i > p_j$, and $(a - p_i)/2$ when $p_i = p_j$. Suppose also that there are no fixed costs and that marginal costs are constant at c, where $c < a$. Show that if the firms choose prices simultaneously, then the unique Nash equilibrium is that both firms charge the price c.

Resolution:

Exercise 18. *Market Entry* [Exam level; difficulty 4/5]

Source. Tadelis (2013), *Game Theory: An Introduction*, Ch. 6 "Mixed Strategies", Exercise 6.8 "Market Entry", p. 125 [PDF 144]. No answer in the book. Reproduced verbatim.

6.8 Market Entry: Three firms are considering entering a new market. The payoff for each firm that enters is $\frac{150}{n}$, where n is the number of firms that enter. The cost of entering is 62.

a. Find all the pure-strategy Nash equilibria.

b. Find the symmetric mixed-strategy equilibrium in which all three players enter with the same probability.

Resolution:

Exercise 19. *A unique IESDS survivor is a Nash equilibrium* [Advanced; difficulty 4/5]

Source. Mas-Colell, Whinston & Green (1995), *Microeconomic Theory*, Ch. 8 "Simultaneous-Move Games", Exercises, Exercise 8.D.2^{B}, p. 263 [PDF 142, right half]. Answer in the MWG solutions manual (`Livros/`). Reproduced verbatim.

8.D.2^{B} Show that if there is a unique profile of strategies that survives iterated removal of strictly dominated strategies, this profile is a Nash equilibrium.

Resolution:

Exercise 20. *Existence in 2×2 games* [Advanced; difficulty 5/5]

Source. Mas-Colell, Whinston & Green (1995), *Microeconomic Theory*, Ch. 8 "Simultaneous-Move Games", Exercises, Exercise 8.D.6^{B}, p. 264 [PDF 143, left half]. Answer in the MWG solutions manual (`Livros/`; 8.D.6 is there). Reproduced verbatim.

8.D.6^{B} Consider any two-player game of the following form (where letters indicate arbitrary payoffs):

		Player 2	
		b_1	b_2
Player 1	a_1	u, v	ℓ, m
	a_2	w, x	y, z

Show that a mixed strategy Nash equilibrium always exists in this game. [*Hint:* Define player 1's strategy to be his probability of choosing action a_1 and player 2's to be his probability of choosing b_1; then examine the best-response correspondences of the two players.]

Resolution:

Block 3 — Dynamic games: extensive form, backward induction, SPE *(Lecture 3)*

Exercise 21. *Strategies* [Foundations; difficulty 1/5]

Source. Tadelis (2013), *Game Theory: An Introduction*, Ch. 7 "Preliminaries", Exercise 7.1 "Strategies", p. 147 [PDF 166]. No answer in the book. Reproduced verbatim.

7.1 Strategies: Imagine an extensive-form game in which player i has K information sets.

a. If the player has an identical number of m possible actions in each information set, how many pure strategies does he have?

b. If the player has m_k actions in information set $k \in \{1, 2, \ldots, K\}$, how many pure strategies does the player have?

Resolution:

Exercise 22. *Sequential Version of the Prisoner's Dilemma Game* [Foundations; difficulty 1/5]

Source. Muñoz-Garcia & Toro-Gonzalez (2019), *Strategy and Game Theory: Practice Exercises with Answers*, 2nd ed., Ch. 4 "Sequential-Move Games with Complete Information", Exercise 3 "Sequential Version of the Prisoner's Dilemma Game"[A], p. 160 [PDF 172]. Worked answer in the book, right after the statement. Reproduced verbatim (statement only); Fig. 4.9 cropped from the page.

Consider the game tree in Fig. 4.9, representing the sequential-move version of the Prisoner's dilemma game, where player 1 acts first, and observing its move player 2 responds.

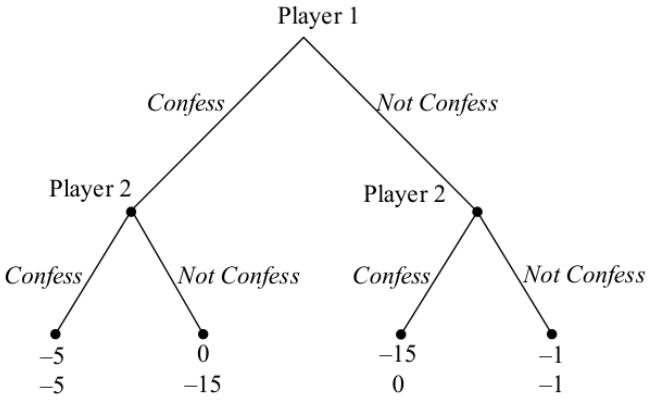


Fig. 4.9 Prisoner's dilemma sequential-move game

(a) Find the best responses of the second mover, for each of the first mover's actions.

(b) Find the first mover's equilibrium action and describe the subgame perfect Nash equilibrium of the game.

(c) Does the equilibrium behavior in the sequential-move game differ from its simultaneous-move version?

Referenced material. Source. "its simultaneous-move version": Muñoz-Garcia & Toro-Gonzalez (2019), Ch. 1, Exercise 5 "Prisoners' Dilemma Game", Fig. 1.8, p. 9 [PDF 22]. Matrix rebuilt from the page.

		Prisoner 2: Confess	Prisoner 2: Not confess
Prisoner 1	Confess	-5, -5	0, -15
	Not confess	-15, 0	-1, -1

Fig. 1.8 Normal-form of Prisoners' dilemma game

Resolution:

Exercise 23. *Strategies and Equilibrium* [Foundations; difficulty 2/5]

Source. Tadelis (2013), *Game Theory: An Introduction*, Ch. 7 "Preliminaries", Exercise 7.2 "Strategies and Equilibrium", p. 147 [PDF 166]. No answer in the book. Reproduced verbatim.

7.2 Strategies and Equilibrium: Consider a two-player game in which player 1 can choose A or B. The game ends if he chooses A while it continues to player 2 if he chooses B. Player 2 can then choose C or D, with the game ending after C and continuing again with player 1 after D. Player 1 can then choose E or F, and the game ends after each of these choices.

a. Model this as an extensive-form game tree. Is it a game of perfect or imperfect information?

b. How many terminal nodes does the game have? How many information sets?

c. How many pure strategies does each player have?

d. Imagine that the payoffs following choice A by player 1 are $(2, 0)$, those following C by player 2 are $(3, 1)$, those following E by player 1 are $(0, 0)$, and those following F by player 1 are $(1, 2)$. What are the Nash equilibria of this game? Does one strike you as more "appealing" than the other? If so, explain why.

Resolution:

Exercise 24. *Technology Adoption* [Exam level; difficulty 3/5]

Source. Tadelis (2013), *Game Theory: An Introduction*, Ch. 8 "Credibility and Sequential Rationality", Exercise 8.5 "Technology Adoption", p. 170 [PDF 189]. No answer in the book. Reproduced verbatim.

8.5 Technology Adoption: During the adoption of a new technology a CEO (player 1) can design a new task for a division manager. The new task can be either high level (H) or low level (L). The manager simultaneously chooses to invest in good training (G) or bad training (B). The payoffs from this interaction are given by the following matrix:

		Player 2	
		G	B
Player 1	H	5, 4	-5, 2
	L	2, -2	0, 0

a. Present the game in extensive form (a game tree) and solve for all the Nash equilibria and subgame-perfect equilibria.

b. Now assume that before the game is played the CEO can choose not to adopt this new technology, in which case the payoffs are (1, 1), or to adopt it, in which case the game is played. Present the *entire* game in extensive form. How many proper subgames does it have?

c. Solve for all the Nash equilibria and subgame-perfect equilibria of the game described in (b).

Resolution:

Exercise 25. *Playing It Safe* [Exam level; difficulty 3/5]

Source. Tadelis (2013), *Game Theory: An Introduction*, Ch. 8 "Credibility and Sequential Rationality", Exercise 8.10 "Playing It Safe", p. 172 [PDF 191]. No answer in the book. Reproduced verbatim.

8.10 Playing It Safe: Consider the following dynamic game: Player 1 can choose to play it safe (denote this choice by S), in which case both he and player 2 get a payoff of 3 each, or he can risk playing a game with player 2 (denote this choice by R). If he chooses R then they play the following simultaneous-move game:

		Player 2	
		A	B
Player 1	C	8, 0	0, 2
	D	6, 6	2, 2

a. Draw a game tree that represents this game. How many proper subgames does it have?

b. Are there other game trees that would work? Explain briefly.

c. Construct the matrix representation of the normal form of this dynamic game.

d. Find all the Nash and subgame-perfect equilibria of the dynamic game.

Resolution:

Exercise 26. *Centipedes* [Exam level; difficulty 3/5]

Source. Tadelis (2013), *Game Theory: An Introduction*, Ch. 7 "Preliminaries", Exercise 7.4 "Centipedes", p. 148 [PDF 167]. No answer in the book. Reproduced verbatim.

7.4 Centipedes: Imagine a two-player game that proceeds as follows. A pot of money is created with \$6 in it initially. Player 1 moves first, then player 2, then player 1 again, and finally player 2 again. At each player's turn to move, he has two possible actions: grab (G) or share (S). If he grabs he gets $\frac{2}{3}$ of the current pot of money, the other player gets $\frac{1}{3}$ of the pot, and the game ends. If he shares then the size of the current pot is multiplied by $\frac{3}{2}$ and the next player gets to move. At the last stage at which player 2 moves, if he chooses S then the pot is still multiplied by $\frac{3}{2}$, player 2 gets $\frac{1}{3}$ of the pot, and player 1 gets $\frac{2}{3}$ of the pot.

a. Model this as an extensive-form game tree. Is it a game of perfect or imperfect information?

b. How many terminal nodes does the game have? How many information sets?

c. How many pure strategies does each player have?

d. Find the Nash equilibria of this game. How many outcomes can be supported in equilibrium?

e. Now imagine that at the last stage at which player 2 moves, if he chooses to share then the pot is equally split among the players. Does your answer to part (d) change?

Resolution:

Exercise 27. *The Dean's Dilemma* [Exam level; difficulty 4/5]

Source. Tadelis (2013), *Game Theory: An Introduction*, Ch. 7 "Preliminaries", Exercise 7.9 "The Dean's Dilemma", pp. 149–150 [PDF 168–169]. No answer in the book. Reproduced verbatim.

7.9 The Dean's Dilemma: A student stole the DVD player from the student lounge. The dean of students (player 1) suspects the student (player 2) and begins collecting evidence. However, evidence collection is a random process, and concrete evidence will be available to the dean only with probability $\frac{1}{2}$. The student knows that the evidence-gathering process is under way but does not know whether the dean has collected evidence or not. The game proceeds as follows: The dean realizes whether or not he has evidence and can then choose his action, whether to accuse the student (A) or bounce the case (B) and forget it. Once accused, the student has two options: he can either confess (C) or deny (D). Payoffs are realized as follows: If the dean bounces the case then both players get 0. If the dean accuses the student and the student confesses, the dean gains 2 and the student loses 2. If the dean accuses the student and the student denies, then payoffs depend on the evidence: If the dean has no evidence then he loses face, which gives him a loss of 4, while the student gains glory, which gives him a payoff of 4. If, however, the dean has evidence then he is triumphant and gains 4, while the student is put on probation and loses 4.

a. Draw the game tree that represents the extensive form of this game.

b. Write down the matrix that represents the normal form of the extensive form you drew in (a).

c. Solve for the Nash equilibria of the game.

Resolution:

Exercise 28. *Perfect and Imperfect Recall* [Exam level; difficulty 4/5]

Source. Tadelis (2013), *Game Theory: An Introduction*, Ch. 7 "Preliminaries", Exercise 7.10 "Perfect and Imperfect Recall", p. 150 [PDF 169]. No answer in the book. Reproduced verbatim; Figure 7.14 cropped from the page.

7.10 Perfect and Imperfect Recall: Consider the game depicted in Figure 7.14.

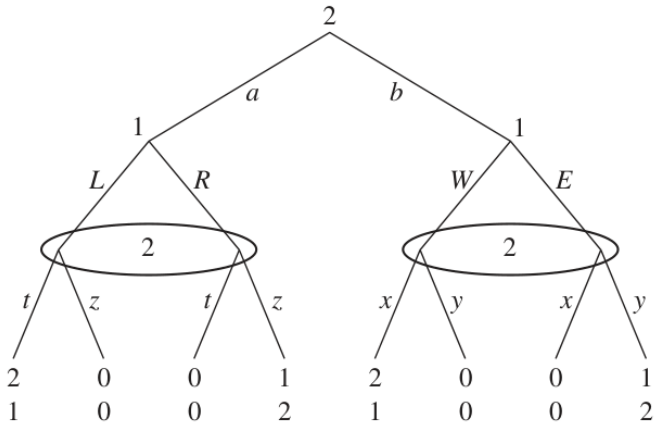


FIGURE 7.14 Exercise 7.10.

a. What are the pure-strategy sets for each player?

b. Show that for any behavioral strategy for player 1 there is a mixed strategy that leads to the same distribution over the terminal nodes regardless of the strategy chosen by player 2.

c. Show that for any behavioral strategy for player 2 there is a mixed strategy that leads to the same distribution over the terminal nodes regardless of the strategy chosen by player 1.

d. Now imagine that the game does not have perfect recall so that player 2's bottom two information sets are now one large information set. Can you find an example showing that the claim in (c) is no longer true?

Resolution:

Exercise 29. *Zero-sum games of perfect information* [Advanced; difficulty 4/5]

Source. Mas-Colell, Whinston & Green (1995), *Microeconomic Theory*, Ch. 9 "Dynamic Games", Exercises, Exercise 9.B.4^{B}, p. 302 [PDF 162, left half]. Answer in the MWG solutions manual (`Livros/`). Reproduced verbatim.

9.B.4^{B} Prove that in a finite *zero-sum* game of perfect information, there are unique subgame perfect Nash equilibrium payoffs.

Resolution:

Exercise 30. *The Rotten Kid Theorem* [Advanced; difficulty 5/5]

Source. Gibbons (1992), *Game Theory for Applied Economists*, Ch. 2 "Dynamic Games of Complete Information", §2.6 Problems (Section 2.1), Problem 2.1, p. 130 [PDF 137]. No answer in the book. Reproduced verbatim.

2.1. Suppose a parent and child play the following game, first analyzed by Becker (1974). First, the child takes an action, A, that produces income for the child, $I_C(A)$, and income for the parent, $I_P(A)$. (Think of $I_C(A)$ as the child's income net of any costs of the action A.) Second, the parent observes the incomes I_C and I_P and then chooses a bequest, B, to leave to the child. The child's payoff is $U(I_C + B)$; the parent's is $V(I_P - B) + kU(I_C + B)$, where $k > 0$ reflects the parent's concern for the child's well-being. Assume that: the action is a nonnegative number, $A \geq 0$; the income functions $I_C(A)$ and $I_P(A)$ are strictly concave and are maximized at $A_C > 0$ and $A_P > 0$, respectively; the bequest B can be positive or negative; and the utility functions U and V are increasing and strictly concave. Prove the "Rotten Kid" Theorem: in the backwards-induction outcome, the child chooses the action that maximizes the family's aggregate income, $I_C(A) + I_P(A)$, even though only the parent's payoff exhibits altruism.

Resolution:

Block 4 — Credible threats and subgame perfection *(Lecture 4)*

Exercise 31. *Repeated-game definitions* [Foundations; difficulty 1/5]

Source. Gibbons (1992), *Game Theory for Applied Economists*, Ch. 2 "Dynamic Games of Complete Information", §2.6 Problems (Section 2.3), Problem 2.12, p. 135 [PDF 142]. No answer in the book. Reproduced verbatim.

2.12. What is a strategy in a repeated game? What is a subgame in a repeated game? What is a subgame-perfect Nash equilibrium?

Resolution:

Exercise 32. *Entering an Industry* [Foundations; difficulty 1/5]

Source. Tadelis (2013), *Game Theory: An Introduction*, Ch. 7 "Preliminaries", Exercise 7.6 "Entering an Industry", pp. 148–149 [PDF 167–168]. No answer in the book. Reproduced verbatim.

7.6 Entering an Industry: A firm (player 1) is considering entering an established industry with one incumbent firm (player 2). Player 1 must choose whether or not to enter the industry. If player 1 enters the industry then player 2 can either accommodate the entry or fight the entry by waging a price war. Player 1's most-preferred outcome is entering with player 2 not fighting, and its least-preferred outcome is entering with player 2 fighting. Player 2's most-preferred outcome is player 1 not entering, and its least-preferred outcome is player 1 entering with player 2 fighting.

a. Model this as an extensive-form game tree (choose payoffs that represent the preferences).

b. How many pure strategies does each player have?

c. Find all the Nash equilibria of this game.

Resolution:

Exercise 33. *Entry Deterrence 1* [Foundations; difficulty 2/5]

Source. Tadelis (2013), *Game Theory: An Introduction*, Ch. 8 "Credibility and Sequential Rationality", Exercise 8.8 "Entry Deterrence 1", pp. 171–172 [PDF 190–191]. No answer in the book. Reproduced verbatim.

8.8 Entry Deterrence 1: NSG Corporation is considering entry into the local phone market in the San Francisco Bay Area. The incumbent, S&P Corporation, predicts that a price war will result if NSG enters. If NSG stays out, S&P earns monopoly profits valued at \$10 million (net present value of profits) while NSG earns zero. If NSG enters, it must incur irreversible entry costs of \$2 million. If there is a price war, each firm earns \$1 million. S&P always has the option of accommodating entry (i.e., not starting a price war). In such a case both firms earn \$4 million (net present value). Suppose that the timing is such that NSG first has to choose whether or not to enter the market. Then S&P decides whether to "accommodate entry" or "engage in a price war." What is the subgame-perfect equilibrium outcome to this sequential game? Set up a game tree.

Resolution:

Exercise 34. *Investment in the Future* [Exam level; difficulty 3/5]

Source. Tadelis (2013), *Game Theory: An Introduction*, Ch. 8 "Credibility and Sequential Rationality", Exercise 8.6 "Investment in the Future", pp. 170–171 [PDF 189–190]. No answer in the book. Reproduced verbatim.

8.6 Investment in the Future: Consider two firms that play a Cournot competition game with demand $p = 100 - q$ and costs for each firm given by $c_i(q_i) = 10q_i$. Imagine that before the two firms play the Cournot game firm 1 can invest in cost reduction. If it invests the costs of firm 1 will drop to $c_1(q_1) = 5q_1$. The cost of investment is $F > 0$. Firm 2 does not have this investment opportunity.

a. Find the value F^* for which the unique subgame-perfect equilibrium involves firm 1 investing.

b. Assume that $F > F^*$. Find a Nash equilibrium of the game that is not subgame perfect.

Resolution:

Exercise 35. *Stackelberg with Two Firms* [Exam level; difficulty 3/5]

Source. Muñoz-Garcia & Toro-Gonzalez (2019), *Strategy and Game Theory: Practice Exercises with Answers*, 2nd ed., Ch. 4 "Sequential-Move Games with Complete Information", Exercise 11 "Stackelberg with Two Firms"[A], p. 180 [PDF 192]. Worked answer in the book, right after the statement. Reproduced verbatim (statement only).

Consider a leader and a follower in a Stackelberg game of quantity competition. Firms face an inverse demand curve $p(Q) = 1 - Q$, where $Q = q_L + q_F$ denotes aggregate output. The leader faces a constant marginal cost $c_L > 0$ while the follower's constant marginal cost is $c_F > 0$, where $1 > c_F \geq c_L$, indicating that the leader has access to a more efficient technology than the follower.

Part (a) Find the follower's best response function, BRF_F, i.e., $q_F(q_L)$.

Part (b) Determine each firm's output strategy in the subgame perfect equilibrium (SPNE) of this sequential-move game.

Part (c) Under which conditions on c_L can you guarantee that both firms produce strictly positive output levels?

Part (d) Assuming cost symmetry, i.e., $c_L = c_F = c$, determine the aggregate equilibrium output level in the Stackelberg game, and its associated equilibrium price. Then compare them with those arising in (1) a monopoly; in (2) the Cournot game of quantity competition; and (3) in the Bertrand game of price competition (which, under cost symmetry, is analogous to a perfectly competitive industry) [*Hint*: For a more direct and visual comparison, depict the inverse demand curve $p(Q) = 1 - Q$, and locate each of these four aggregate output levels in the horizontal axis (with its associated prices in the vertical axis).].

Resolution:

Exercise 36. *Resident Assistant Selection with a Twist* [Exam level; difficulty 3/5]

Source. Tadelis (2013), *Game Theory: An Introduction*, Ch. 8 "Credibility and Sequential Rationality", Exercise 8.11 "Resident Assistant Selection with a Twist", pp. 172–173 [PDF 191–192]. No answer in the book. Reproduced verbatim.

8.11 Resident Assistant Selection with a Twist: Two staff managers in the ΠΒΦ sorority, the house manager (player 1) and kitchen manager (player 2), must select a resident assistant from a pool of three candidates: $\{a, b, c\}$. Player 1 prefers a to b, and b to c. Player 2 prefers b to a, and a to c. The process that is imposed on them is as follows: First, the house manager vetoes one of the candidates and announces the veto to the central office for staff selection and to the kitchen manager. Next the kitchen manager vetoes one of the remaining two candidates and announces it to the central office. Finally the director of the central office assigns the remaining candidate to be a resident assistant at ΠΒΦ.

a. Model this as an extensive-form game (using a game tree) in which a player's most-preferred candidate gives a payoff of 2, the second most-preferred candidate gives a payoff of 1, and the last candidate gives 0.

b. Find the subgame-perfect equilibrium for this game. Is it unique?

c. Are there Nash equilibria that are not subgame-perfect equilibria?

d. Now assume that before the two players play the game, player 2 can send an alienating e-mail

to one of the candidates, which would result in that candidate withdrawing her application. Would player 2 choose to do this, and if so, with which candidate?

Resolution:

Exercise 37. *Industry Leader* [Exam level; difficulty 4/5]

Source. Tadelis (2013), *Game Theory: An Introduction*, Ch. 8 "Credibility and Sequential Rationality", Exercise 8.4 "Industry Leader", p. 170 [PDF 189]. No answer in the book. Reproduced verbatim. (Gibbons Problem 2.6, p. 133, poses the same game and asks only for the subgame-perfect outcome; it is not repeated in this list.)

8.4 Industry Leader: Three oligopolists operate in a market with inverse demand given by $P(Q) = a - Q$, where $Q = q_1 + q_2 + q_3$, and q_i is the quantity produced by firm i. Each firm has a constant marginal cost of production, c, and no fixed cost. The firms choose their quantities dynamically as follows: Firm 1, which is the industry leader, chooses $q_1 \geq 0$. Firms 2 and 3 observe q_1 and then simultaneously choose q_2 and q_3, respectively.

a. How many proper subgames does this dynamic game have? Explain briefly.

b. Is it a game of perfect or imperfect information? Explain briefly.

c. What is the subgame-perfect equilibrium of this game? Show that it is unique.

d. Find a Nash equilibrium that is not a subgame-perfect equilibrium.

Resolution:

Exercise 38. *Junk Mail Advertising* [Exam level; difficulty 4/5]

Source. Tadelis (2013), *Game Theory: An Introduction*, Ch. 8 "Credibility and Sequential Rationality", Exercise 8.13 "Junk Mail Advertising", pp. 173–174 [PDF 192–193]. No answer in the book. Reproduced verbatim.

8.13 Junk Mail Advertising: Suppose a single good is owned by a single seller who values it at $c > 0$ (he can consume the good and get a payoff of c). There is a single buyer who has a small transportation cost $k > 0$ to get to and back from the seller's store, and he values the good at $v > c + k$. The buyer first decides whether to make the commute or stay at home, not buy the good, and receive a payoff of 0. If the buyer commutes to the store, the seller can then make the buyer a take-it-or-leave-it price offer $p \geq 0$. The buyer can then accept the offer, pay p, and get the good, or he can walk out and not buy the good. Assume that c, v, and k are common knowledge.

a. As best you can, draw the extensive form of this game. What is the best response of the buyer at the node at which he decides whether to accept or reject the seller's offer?

b. Find the subgame-perfect equilibrium of the game and show that it is unique. Is it Pareto optimal?

c. Are there Nash equilibria that yield a higher payoff to both players as compared to the subgame-perfect equilibrium you found in (b)?

d. Now assume that before the game is played the seller can, at a small cost $\varepsilon < (v - c - k)$, send the buyer a postcard that commits the seller to a certain price at which the buyer can buy the good (e.g., "bring this coupon and get the good at a price p"). Would the seller choose to do so? Justify your answer with an equilibrium analysis.

Resolution:

Exercise 39. *Debt and Repayment* [Advanced; difficulty 4/5]

Source. Tadelis (2013), *Game Theory: An Introduction*, Ch. 8 "Credibility and Sequential Rationality", Exercise 8.7 "Debt and Repayment", p. 171 [PDF 190]. No answer in the book. Reproduced verbatim.

8.7 Debt and Repayment: A project costing \$100 yields a gross return of \$110. A lender (player 1) is approached by a debtor (player 2) requesting a standard loan contract to complete the project. If the lender chooses *not to offer* a loan, then both parties earn nothing. If the lender chooses to *offer* a loan of \$100, the debtor can realize the project's gains and is obliged by contract to repay \$105. For simplicity assume that money is continuous and that the debtor can choose to return any amount of money $x \leq 110$. Ignore the time value of money. Assume first that no legal system is in place that can compel the debtor to repay, so that default on the loan (less than full repayment) carries no repercussions for the debtor.

a. Model this as an extensive-form game tree as best you can and find a subgame-perfect equilibrium of this game. Is it unique?

b. Now assume that there is a legal system in place that allows the lender to *voluntarily* choose whether or not to sue when the debtor defaults and repays an amount $x < 105$. Furthermore assume that it is *costless* to use the legal system (it is supplied by the state), and if the lender sues a debtor that defaulted, the lender will get the \$105 repaid in full. After paying the lender, the borrower will pay a fine of \$5 to the court, above and beyond the repayment. Model this as

an extensive-form game tree as best you can and find a subgame-perfect equilibrium of this game. Is it unique?

c. Are there Nash equilibria in the game described in (b) that are not subgame-perfect equilibria?

d. Now assume that using the legal system is *costly:* if the lender sues, he pays lawyers a *legal fee* of \$105 (this is the lawyers' price; it is unrelated to the contract). The rest of the game proceeds as before. (If the lender sues a debtor that defaulted, the lender will get repaid in full; after paying the lender, the borrower will pay a fine of \$5 above and beyond the repayment.) Model this as an extensive-form game tree as best you can and find a subgame-perfect equilibrium of this game. Is it unique?

e. Are there Nash equilibria in the game described in (d) that are not subgame-perfect equilibria?

f. Now assume that a change in the law is proposed: upon default, if a debtor is sued he has to first repay the lender \$105 and then pay the legal fees of \$105 above and beyond repayment of the loan, and no extra fine is imposed. Should the lender be willing to pay to have this change in the law enacted? If so, how much?

g. If you were a social planner, would you implement the suggested law?

Resolution:

Exercise 40. *Observation, multiple SPNE and moving first* [Advanced; difficulty 5/5]

Source. Mas-Colell, Whinston & Green (1995), *Microeconomic Theory*, Ch. 9 "Dynamic Games", Exercises, Exercise 9.B.5^{B} (E. Maskin), p. 302 [PDF 162, left half]. Answer in the MWG solutions manual (`Livros/`; 9.B.5 is there). Reproduced verbatim; the label of the last part is blurred in the scan and is read as (e), as in the solutions manual.

9.B.5^{B} (E. Maskin) Consider a game with two players, player 1 and player 2, in which each player i can choose an action from a finite set M_i that contains m_i actions. Player i's payoff if the action choices are (m_1, m_2) is $\phi_i(m_1, m_2)$.

(a) Suppose, first, that the two players move simultaneously. How many strategies does each player have?

(b) Now suppose that player 1 moves first and that player 2 observes player 1's move before choosing her move. How many strategies does each player have?

(c) Suppose that the game in (b) has multiple SPNEs. Show that if this is the case, then there exist two pairs of moves (m_1, m_2) and (m_1', m_2') (where either $m_1 \neq m_1'$ or $m_2 \neq m_2'$) such that either

$$\text{(i)} \quad \phi_1(m_1, m_2) = \phi_1(m_1', m_2')$$

or

$$\text{(ii)} \quad \phi_2(m_1, m_2) = \phi_2(m_1', m_2').$$

(d) Suppose that for any two pairs of moves (m_1, m_2) and (m_1', m_2') such that $m_1 \neq m_1'$ or $m_2 \neq m_2'$, condition (ii) is violated (i.e., player 2 is never indifferent between pairs of moves). Suppose also that there exists a pure strategy Nash equilibrium in the game in **(a)** in which π_1 is player 1's payoff. Show that in any SPNE of the game in **(b)**, player 1's payoff is at least π_1. Would this conclusion necessarily hold for any *Nash* equilibrium of the game in **(b)**?

(e) Show by example that the conclusion in **(d)** may fail either if condition (ii) holds for some strategy pairs (m_1, m_2), (m_1', m_2') with $m_1 \neq m_1'$ or $m_2 \neq m_2'$ or if we replace the phrase *pure strategy Nash equilibrium* with the phrase *mixed strategy Nash equilibrium*.

Resolution:

Block 5 — Repeated games and bargaining *(Lecture 5)*

Exercise 41. *Finitely repeated game* [Foundations; difficulty 1/5]

Source. Cabral (2017), *Introduction to Industrial Organization*, 2nd ed., Ch. 7 "Games and Strategies", Exercises, Exercise 7.13 "Finitely repeated game", p. 182 [PDF 194]. No answer in the book. Reproduced verbatim.

7.13. Finitely repeated game. Suppose that the game depicted in Figure 7.1 is repeated T times, where T is known. Show that the only subgame perfect equilibrium is for players to choose (B, R) in every period.

Referenced material. Source. "the game depicted in Figure 7.1": Cabral (2017), Ch. 7, Figure 7.1 "The 'prisoner's dilemma' game" and the sentence that explains how its cells are read, p. 160 [PDF 172]. Figure cropped from the page; text reproduced verbatim.

For each combination of strategies by each player, the respective matrix cell shows the payoffs received by each player. In the lower left corner, the payoff received by Player 1; in the top right corner, the payoff received by Player 2.

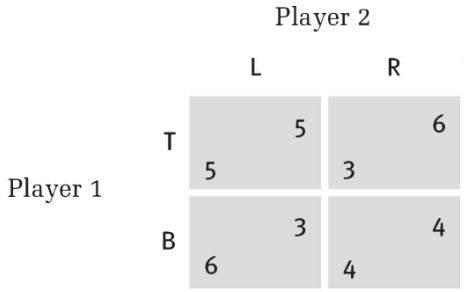


FIGURE 7.1
The "prisoner's dilemma" game

Resolution:

Exercise 42. *Grim Trigger* [Foundations; difficulty 2/5]

Source. Tadelis (2013), *Game Theory: An Introduction*, Ch. 10 "Repeated Games", Exercise 10.2 "Grim Trigger", p. 215 [PDF 234]. No answer in the book. Reproduced verbatim.

10.2 Grim Trigger: Consider the infinitely repeated game with discount factor $\delta < 1$ of the following variant of the Prisoner's Dilemma:

		Player 2		
		L	C	R
Player 1	T	6, 6	-1, 7	-2, 8
	M	7, -1	4, 4	-1, 5
	B	8, -2	5, -1	0, 0

a. For which values of the discount factor δ can the players support the pair of actions (M, C) played in every period?

b. For which values of the discount factor δ can the players support the pair of actions (T, L) played in every period? Why is your answer different from that for (a)?

Resolution:

Exercise 43. *Infinitely Repeated Prisoner's Dilemma Game* [Foundations; difficulty 2/5]

Source. Muñoz-Garcia & Toro-Gonzalez (2019), *Strategy and Game Theory: Practice Exercises with Answers*, 2nd ed., Ch. 6 "Repeated Games and Correlated Equilibria", Exercise 1 "Infinitely Repeated Prisoner's Dilemma Game"[A], p. 274 [PDF 285]. Worked answer in the book, right after the statement. Reproduced verbatim (statement only); Table 6.1 rebuilt from the same page.

Consider the infinitely repeated version of the following prisoner's dilemma game, where C denotes confess and NC represents not confess (Table 6.1):

Part (a) Can players support the cooperative outcome (NC, NC) as a SPNE by using tit-for-tat strategies, whereby players punish deviations from NC in a past period by reverting to the stage-game Nash equilibrium where they both confess (C, C) for just one period, and then returning to cooperation?

Part (b) Consider now that players play tit-for-tat, but reverting to the stage-game Nash equilibrium where they both confess (C, C) for two periods, and then players return to cooperation. Can the cooperative outcome (NC, NC) be supported as a SPNE?

Part (c) Suppose that players use strategies that punish deviation from cooperation by reverting to the stage-game Nash equilibrium for ten periods before returning to cooperation. Compute the threshold discount factor, δ, above which cooperation can be sustained as a SPNE of the infinitely repeated game.

		Player 2	
		C	NC
Player 1	C	2, 2	6, 0
	NC	0, 6	3, 3

Table 6.1 Prisoner's dilemma game

Resolution:

Exercise 44. *Representing Feasible, Individually Rational Payoffs* [Exam level; difficulty 3/5]

Source. Muñoz-Garcia & Toro-Gonzalez (2019), *Strategy and Game Theory: Practice Exercises with Answers*, 2nd ed., Ch. 6 "Repeated Games and Correlated Equilibria", Exercise 9 "Representing Feasible, Individually Rational Payoffs"[C], p. 305 [PDF 316]. Worked answer in the book, right after the statement. Reproduced verbatim (statement only); Table 6.6, printed inside the answer on the same page, rebuilt.

Consider an infinitely-repeated game where the stage game is depicted Table 6.6. Player 1 chooses rows and player 2 chooses columns. Players discount the future using a common discount factor δ.

Part (a) What outcomes in the stage-game are consistent with Nash equilibrium play?

Part (b) Let v_1 and v_2 be the repeated game payoffs to player 1 and player 2 respectively. Draw the set of feasible payoffs from the repeated game, explaining any normalization you use.

Part (c) Find the set of individually rational feasible set of payoffs.

Part (d) Find a Nash equilibrium in which the players obtain (9, 9) each period. What restrictions on δ are necessary?

		Player 2	
		L	R
Player 1	U	9, 9	1, 10
	D	10, 1	7, 7

Table 6.6 Simultaneous-move game

Resolution:

Exercise 45. *Not-So-Grim Trigger* [Exam level; difficulty 3/5]

Source. Tadelis (2013), *Game Theory: An Introduction*, Ch. 10 "Repeated Games", Exercise 10.3 "Not-So-Grim Trigger", pp. 215–216 [PDF 234–235]. No answer in the book. Reproduced verbatim, with the book's footnote 12 (p. 216). Here T is the length of the punishment.

10.3 Not-So-Grim Trigger: Consider the infinitely repeated Prisoner's Dilemma with discount factor $\delta < 1$ described by the following matrix:

		Player 2	
		m	f
Player 1	M	4, 4	-1, 5
	F	5, -1	1, 1

Instead of using grim-trigger strategies to support a pair of actions (a_1, a_2) other than (F, f) as a subgame-perfect equilibrium, assume that the players wish to choose a less draconian punishment called a "length-T punishment" strategy. If there is a deviation from (a_1, a_2) then the players will play (F, f) for T periods and then resume playing (a_1, a_2). Let δ_T be the critical discount factor so that if $\delta > \delta_T$ then the adequately defined strategies will implement the desired path of play with length-T punishment as the threat.

a. Let $T = 1$. What is the critical value δ_1 to support the pair of actions (M, m) played in every period?

b. Let $T = 2$. What is the critical value δ_T to support the pair of actions (M, m) played in every period?[12]

c. Compare the two critical values in (a) and (b). How do they differ and what is the intuition for this?

[12] Hint: You should encounter an equation of the form $a\delta^3 - (a+1)\delta + 1 = 0$, for which it is easy to see that $\delta = 1$ is a root. In this case you know that the equation can be written in the form $(\delta - 1)(a\delta^2 + a\delta - 1) = 0$, and you can solve for the other relevant root of the cubic equation.

Resolution:

Exercise 46. *Multistage Practice* [Exam level; difficulty 4/5]

Source. Tadelis (2013), *Game Theory: An Introduction*, Ch. 9 "Multistage Games", Exercise 9.1 "Multistage Practice", pp. 186–187 [PDF 205–206]. No answer in the book. Reproduced verbatim.

9.1 Multistage Practice: Consider the following simultaneous-move game that is played twice (the players observe the first-period outcome prior to the second-period play):

		Player 2		
		L	C	R
Player 1	T	10, 10	2, 12	0, 13
	M	12, 2	5, 5	0, 0
	B	13, 0	0, 0	1, 1

a. Find all the pure-strategy subgame-perfect equilibria with no discounting ($\delta = 1$). Be precise in defining history-contingent strategies for both players.

b. For each of the equilibria you found in (a), find the smallest discount factor that supports it.

Resolution:

Exercise 47. *Reaching a non-Nash payoff in the first stage* [Exam level; difficulty 4/5]

Source. Gibbons (1992), *Game Theory for Applied Economists*, Ch. 2 "Dynamic Games of Complete Information", §2.6 Problems (Section 2.3), Problem 2.11, p. 135 [PDF 142]. No answer in the book (Muñoz-Garcia & Toro-Gonzalez Ch. 6 Ex. 7, p. 299, works the same game). Reproduced verbatim.

2.11. The simultaneous-move game (below) is played twice, with the outcome of the first stage observed before the second stage begins. There is no discounting. Can the payoff $(4, 4)$ be achieved in the first stage in a pure-strategy subgame-perfect Nash equilibrium? If so, give strategies that do so. If not, prove why not.

	L	C	R
T	3, 1	0, 0	5, 0
M	2, 1	1, 2	3, 1
B	1, 2	0, 1	4, 4

Resolution:

Exercise 48. *Asymmetric Patience 1* [Exam level; difficulty 4/5]

Source. Tadelis (2013), *Game Theory: An Introduction*, Ch. 11 "Strategic Bargaining", Exercise 11.5 "Asymmetric Patience 1", pp. 236–237 [PDF 255–256]. No answer in the book. Reproduced verbatim.

11.5 Asymmetric Patience 1: Consider a three-period sequential (alternating-offer) bargaining model in which two players have to split a pie worth 1 (starting with player 1 making the offer). Now the players have different discount factors, δ_1 and δ_2.

a. Compute the outcome of the unique subgame-perfect equilibrium.

b. Show that when $\delta_1 = \delta_2$ player 1 has an advantage.

c. What conditions on δ_1 and δ_2 give player 2 an advantage? Why?

Referenced material. Source. the alternating-offer bargaining model this exercise modifies ("Now the players have different discount factors"): Tadelis (2013), Ch. 11 "Strategic Bargaining", p. 221 [PDF 240]. Reproduced verbatim (excerpts); with $\delta_1 \neq \delta_2$, player i discounts with δ_i.

In the first round:

- Player 1 offers shares $(x, 1-x)$, where player 1 receives x and player 2 receives $1-x$.
- Player 2 then chooses to accept, causing the game to end with payoffs $v_1 = x$ and $v_2 = 1-x$, or reject, causing the game to move to the second round.

In the second round:

- A share $1-\delta$ of the pie is dissipated (the pie is discounted with discount factor $0 < \delta < 1$).
- Player 2 offers shares $(x, 1-x)$, where player 1 receives x and player 2 receives $1-x$.
- Player 1 chooses to accept, causing the game to end with payoffs $v_1 = \delta x$ and $v_2 = \delta(1-x)$, or reject, causing the game to move to the third round.

In the third round and beyond:

- The game continues in the same way, where following rejection in an odd period player 2 gets to offer in the next even period, and vice versa.
- Each period has further discounting of the pie, so that in period t the total pie is worth δ^{t-1}.

[...] Assume for now that the game ends in some prespecified final period T, which will be reached if the players did not reach an agreement in the $T-1$ periods that preceded T. Furthermore assume that period T is a "hard deadline," so that both players receive a payoff of zero if they fail to reach an agreement in this last period.

Resolution:

Exercise 49. *Diluted Happiness* [Advanced; difficulty 4/5]

Source. Tadelis (2013), *Game Theory: An Introduction*, Ch. 10 "Repeated Games", Exercise 10.7 "Diluted Happiness", pp. 217–218 [PDF 236–237]. No answer in the book. Reproduced verbatim.

10.7 Diluted Happiness: Consider a relationship between a bartender and a customer. The bartender serves bourbon to the customer and chooses $x \in [0, 1]$, which is the proportion of bourbon in the drink served, while $1 - x$ is the proportion of water. The cost of supplying such a drink (standard 4-ounce glass) is cx, where $c > 0$. The customer, without knowing x, decides on whether or not to buy the drink at the market price p. If he buys the drink his payoff is $vx - p$, and the bartender's payoff is $p - cx$. Assume that $v > c$ and all payoffs are common knowledge. If the customer does not buy the drink he gets 0 and the bartender gets $-cx$. Because the customer has some experience, once the drink is bought and he tastes it, he learns the value of x, but this is only after he pays for the drink.

a. Find all the Nash equilibria of this game.

b. Now assume that the customer is visiting town for 10 days, and this "bar game" will be played on each of the 10 evenings that the customer is in town. Assume that each player tries to maximize the (nondiscounted) sum of his stage payoffs. Find all the subgame-perfect equilibria of this game.

c. Now assume that the customer is a local, and the players perceive the game as repeated infinitely many times. Assume that each player tries to maximize the discounted sum of his stage payoffs, where the discount rate is $\delta \in (0, 1)$. What is the range of prices p (expressed in the parameters of the problem) for which there exists a subgame-perfect equilibrium in which every day the bartender chooses $x = 1$ and the customer buys at the price p?

d. For which values of δ (expressed in the parameters of the problem) can the price range that you found in (c) exist?

Resolution:

Exercise 50. *Rubinstein's strategies are subgame-perfect* [Advanced; difficulty 5/5]

Source. Gibbons (1992), *Game Theory for Applied Economists*, Ch. 2 "Dynamic Games of Complete Information", §2.6 Problems (Section 2.4), Problem 2.20, pp. 137–138 [PDF 144–145]. No answer in the book. Reproduced verbatim.

2.20. Consider the following strategies in the infinite-horizon version of Rubinstein's bargaining model. (Recall the notational convention that the offer $(s, 1-s)$ means that Player 1 will get s and Player 2 will get $1-s$, independent of who made the offer.) Let $s^* = 1/(1+\delta)$. Player 1 always offers $(s^*, 1-s^*)$ and accepts an offer $(s, 1-s)$ only if $s \geq \delta s^*$. Player 2 always offers $(1-s^*, s^*)$ and accepts an offer $(s, 1-s)$ only if $1-s \geq \delta s^*$. Show that these strategies are a Nash equilibrium. Show that this equilibrium is subgame-perfect.

Referenced material. Source. "the infinite-horizon version of Rubinstein's bargaining model": Gibbons (1992), §2.1.D "Sequential Bargaining", pp. 68–70 [PDF 75–77]. Reproduced verbatim (excerpts); footnote 7 (on the discount factor as $\delta = 1/(1+r)$) omitted.

Players 1 and 2 are bargaining over one dollar. They alternate in making offers: first player 1 makes a proposal that player 2 can accept or reject; if 2 rejects then 2 makes a proposal that 1 can accept or reject; and so on. Once an offer has been rejected, it ceases to be binding and is irrelevant to the subsequent play of the game. Each offer takes one period, and the players are impatient: they discount payoffs received in later periods by the factor δ per period, where $0 < \delta < 1$.

A more detailed description of the timing of the three-period bargaining game is as follows.

(1a) At the beginning of the first period, player 1 proposes to take a share s_1 of the dollar, leaving $1-s_1$ for player 2.

(1b) Player 2 either accepts the offer (in which case the game ends and the payoffs s_1 to player 1 and $1-s_1$ to player 2 are immediately received) or rejects the offer (in which case play continues to the second period).

(2a) At the beginning of the second period, player 2 proposes that player 1 take a share s_2 of the dollar, leaving $1-s_2$ for player 2. (Note the convention that s_t always goes to player 1, regardless of who made the offer.)

(2b) Player 1 either accepts the offer (in which case the game ends and the payoffs s_2 to player 1 and $1-s_2$ to player 2 are immediately received) or rejects the offer (in which case play continues to the third period).

(3) At the beginning of the third period, player 1 receives a share s of the dollar, leaving $1-s$ for player 2, where $0 < s < 1$.

[...] Now consider the infinite-horizon case. The timing is as described previously, except that the exogenous settlement in step (3) is replaced by an infinite sequence of steps (3a), (3b), (4a), (4b), and so on. Player 1 makes the offer in odd-numbered periods, player 2 in even-numbered; bargaining continues until one player accepts an offer.

Resolution: